

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Introduction to Clinical Nutrition

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

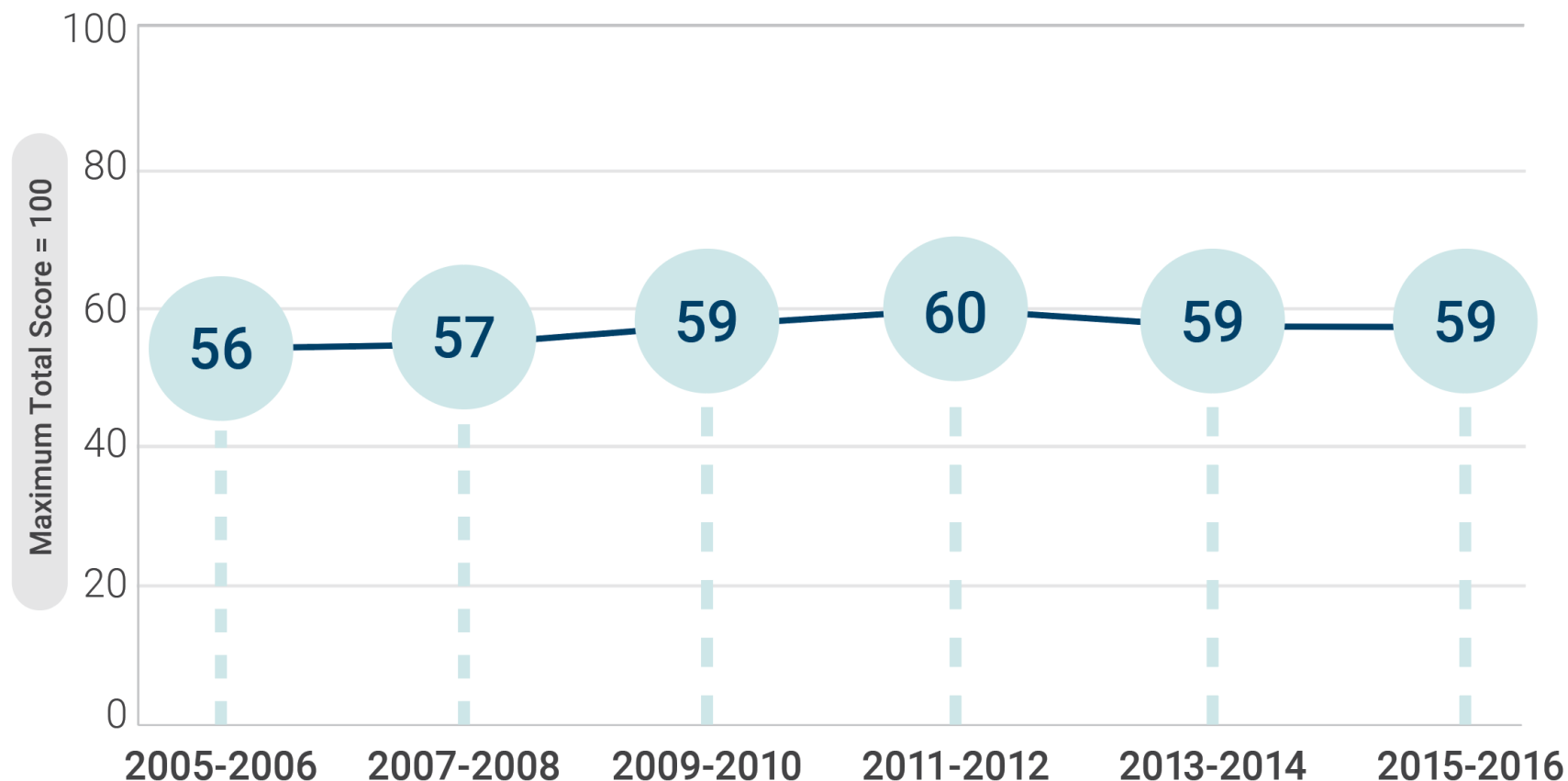
1. Integrate and understand macronutrients, their building blocks, micronutrients, and their recommended intake ranges and diet composition.
2. Integrate and understand calorie requirements and energy balance.
3. Examine and discuss the importance of reading food labels.
4. Analyze the roles of macronutrients and micronutrients in human metabolism and homeostasis.

Nutrition and Health Are Closely Related

“About half of all American adults—117 million individuals—have one or more preventable chronic diseases, many of which are related to poor eating and physical activity patterns.”

Figure I-1

Adherence of the U.S. Population to the *Dietary Guidelines* Over Time, as Measured by the Average Total Healthy Eating Index-2015 Scores

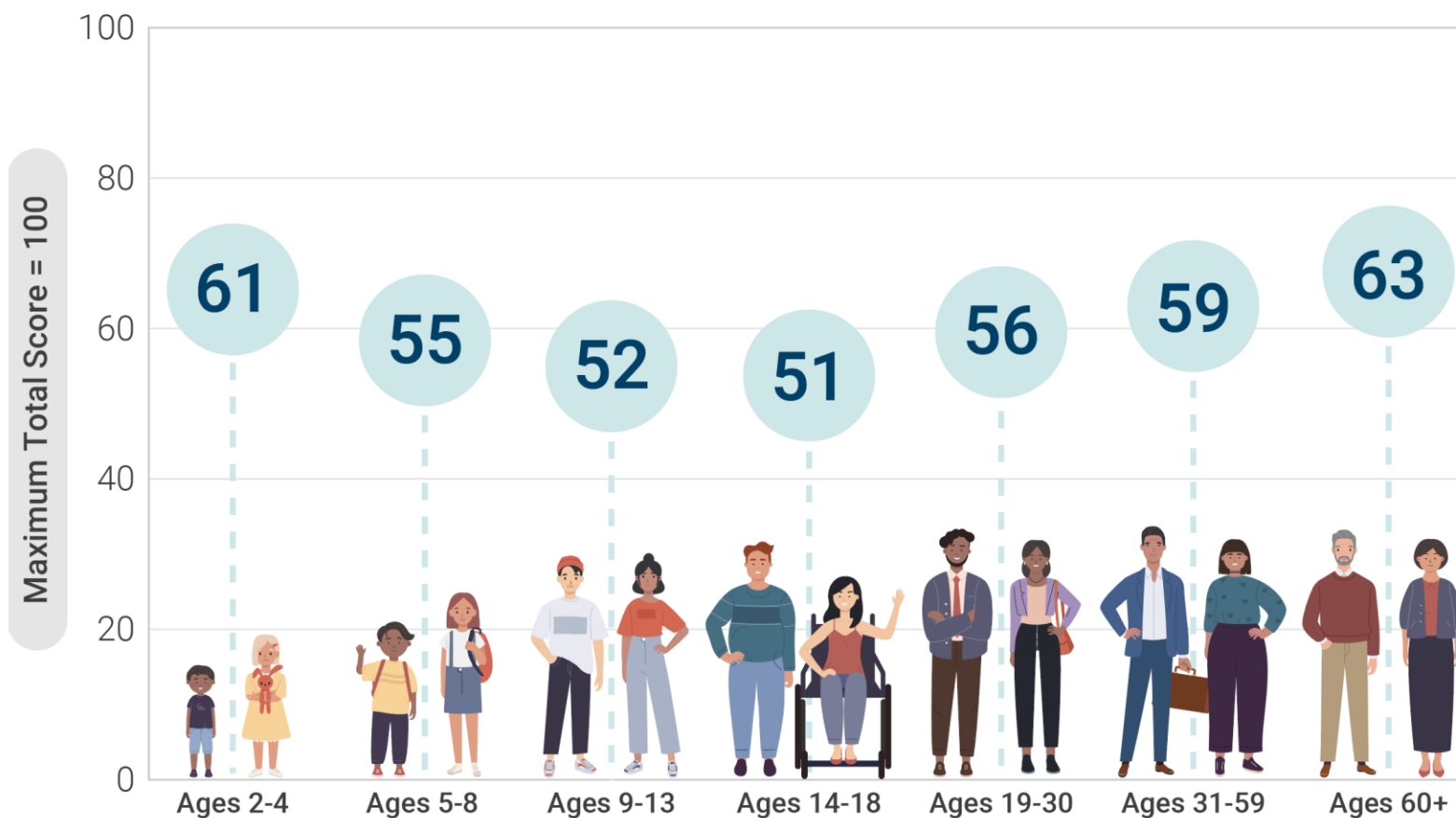


NOTE: HEI-2015 total scores are out of 100 possible points. A score of 100 indicates that recommendations on average were met or exceeded. A higher total score indicates a higher quality diet.

Data Source: Analysis of What We Eat in America, National Health and Nutrition Examination Survey (NHANES) data from 2005-2006 through 2015-2016, ages 2 and older, day 1 dietary intake data, weighted.

Figure 1-4

Adherence of the U.S. Population to the *Dietary Guidelines* Across Life Stages, as Measured by Average Total Healthy Eating Index-2015 Scores



NOTE: HEI-2015 total scores are out of 100 possible points. A score of 100 indicates that recommendations on average were met or exceeded. A higher total score indicates a higher quality diet.

Data Source: Analysis of What We Eat in America, NHANES 2015-2016, ages 2 and older, day 1 dietary intake data, weighted.



Make Every
Bite Count With
the *Dietary
Guidelines*



**Dietary
Guidelines
for Americans**

2020 - 2025



The Guidelines



The Guidelines

Follow a healthy dietary pattern at every life stage.



The Guidelines

Follow a healthy dietary pattern at every life stage.



Customize and enjoy nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations.



2



3



4



1



The Guidelines

Follow a healthy dietary pattern at every life stage.



Customize and enjoy nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations.

1



4



2



3



Focus on meeting food group needs with nutrient-dense foods and beverages, and stay within calorie limits.



The Guidelines





Follow a healthy dietary pattern at every life stage.

- At every life stage—infancy, toddlerhood, childhood, adolescence, adulthood, pregnancy, lactation, and older adulthood—it is never too early or too late to eat healthfully.
- **For about the first 6 months of life**, exclusively feed infants human milk. Continue to feed infants human milk through at least the first year of life, and longer if desired. Feed infants iron-fortified infant formula during the first year of life when human milk is unavailable. Provide infants with supplemental vitamin D beginning soon after birth.
- **At about 6 months**, introduce infants to nutrient-dense complementary foods. Introduce infants to potentially allergenic foods along with other complementary foods. Encourage infants and toddlers to consume a variety of foods from all food groups. Include foods rich in iron and zinc, particularly for infants fed human milk.
- **From 12 months through older adulthood**, follow a healthy dietary pattern across the lifespan to meet nutrient needs, help achieve a healthy body weight, and reduce the risk of chronic disease.

Customize and enjoy nutrient-dense food and beverage choices to reflect personal preferences, cultural traditions, and budgetary considerations.

- A healthy dietary pattern can benefit all individuals regardless of age, race, or ethnicity, or current health status. The *Dietary Guidelines* provides a framework intended to be customized to individual needs and preferences, as well as the foodways of the diverse cultures in the United States.





Focus on meeting food group needs with nutrient-dense foods and beverages, and stay within calorie limits.

An underlying premise of the *Dietary Guidelines* is that nutritional needs should be met primarily from foods and beverages—specifically, nutrient-dense foods and beverages. Nutrient-dense foods provide vitamins, minerals, and other health-promoting components and have no or little added sugars, saturated fat, and sodium.

A healthy dietary pattern consists of nutrient-dense forms of foods and beverages across all food groups, in recommended amounts, and within calorie limits.

The core elements that make up a healthy dietary pattern include:

- **Vegetables of all types**—dark green; red and orange; beans, peas, and lentils; starchy; and other vegetables
- **Fruits**, especially whole fruit
- **Grains**, at least half of which are whole grain
- **Dairy**, including fat-free or low-fat milk, yogurt, and cheese, and/or lactose-free versions and fortified soy beverages and yogurt as alternatives
- **Protein foods**, including lean meats, poultry, and eggs; seafood; beans, peas, and lentils; and nuts, seeds, and soy products
- **Oils**, including vegetable oils and oils in food, such as seafood and nuts

Making Nutrient-Dense Choices: One Meal At a Time



Typical Burrito Bowl
Total Calories = 1,120

White rice (1 ½ cups)

Black beans (⅓ cup)

Chicken cooked with sauce (2 ounces)

No grilled vegetables

Guacamole (½ cup)

Jarred salsa (¼ cup)

Sour cream (¼ cup)

Cheese (⅓ cup)

Jalapeño (5 slices)

Iced tea with sugar (16 ounces)



Nutrient-Dense Burrito Bowl
Total Calories = 715

Brown rice (1 cup) + Romaine lettuce (½ cup)

Black beans, reduced sodium (⅓ cup)

Grilled chicken with spice rub (2 ounces)

Added grilled vegetables (⅓ cup)

Sliced avocado (5 slices)

Fresh salsa/pico de gallo (¼ cup)

No sour cream

Reduced-fat cheese (⅓ cup)

Jalapeño (5 slices)

Iced tea, no sugar (16 ounces)



Limit foods and beverages higher in added sugars, saturated fat, and sodium, and limit alcoholic beverages.

At every life stage, meeting food group recommendations—even with nutrient-dense choices—requires most of a person’s daily calorie needs and sodium limits. A healthy dietary pattern doesn’t have much room for extra added sugars, saturated fat, or sodium—or for alcoholic beverages. A small amount of added sugars, saturated fat, or sodium can be added to nutrient-dense foods and beverages to help meet food group recommendations, but foods and beverages high in these components should be limited.

Limits are:

- **Added sugars**—Less than 10 percent of calories per day starting at age 2. Avoid foods and beverages with added sugars for those younger than age 2.
- **Saturated fat**—Less than 10 percent of calories per day starting at age 2.
- **Sodium**—Less than 2,300 milligrams per day—and even less for children younger than age 14.
- **Alcoholic beverages**—Adults of legal drinking age can choose not to drink or to drink in moderation by limiting intake to 2 drinks or less in a day for men and 1 drink or less in a day for women, when alcohol is consumed. Drinking less is better for health than drinking more. There are some adults who should not drink alcohol, such as women who are pregnant.

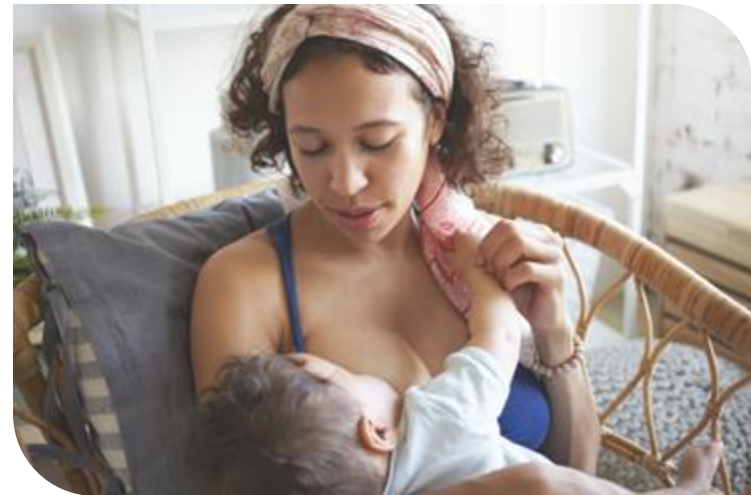
The *Dietary Guidelines for Americans, 2020-2025* includes nearly all of the science-based recommendations of the 2020 Dietary Guidelines Advisory Committee

- The 2020-2025 edition of the *Dietary Guidelines* emphasizes the importance of limiting intakes of added sugars and alcoholic beverages as recommended by the Committee, but does not include changes to quantitative recommendations.
- Any revisions to previous editions of the *Dietary Guidelines* must have sufficient scientific justification, and by law, must be based on the preponderance of scientific and medical knowledge current at the time.
- The preponderance of evidence supports limiting intakes of added sugars and alcoholic beverages to promote health and prevent disease; however, the evidence reviewed since the 2015-2020 edition does not substantiate quantitative changes at this time.
- USDA and HHS encourage more research on the relationship between added sugars and alcoholic beverages and health, and plan to monitor these topics.

Chapter 2. Infants and Toddlers

Key Recommendations

- **For about the first 6 months of life**, exclusively feed infants human milk. Continue to feed infants human milk through at least the first year of life, and longer if desired. Feed infants iron-fortified infant formula during the first year of life when human milk is unavailable.
- Provide infants with supplemental vitamin D beginning soon after birth.



Chapter 2. Infants and Toddlers

Key Recommendations

- **At about 6 months**, introduce infants to nutrient-dense complementary foods.
- Introduce infants to potentially allergenic foods along with other complementary foods.
- Encourage infants and toddlers to consume a variety of foods from all food groups. Include foods rich in iron and zinc, particularly for infants fed human milk.
- Avoid foods and beverages with added sugars.
- Limit foods and beverages higher in sodium.
- As infants wean from human milk or infant formula, transition to a healthy dietary pattern.

Chapters 3-6

Tailored Guidance Across Life Stages

Each chapter includes:

- Healthy U.S.-Style Dietary Pattern at relevant calorie levels
- Overview of current intakes compared to recommendations
- Special dietary considerations
- Resources to support healthy eating

**For lifelong good health, make every bite count
with the *Dietary Guidelines for Americans***



The *Dietary Guidelines for Americans*:

What It Is, What It Is Not

- Provide ***evidence-based recommendations*** about the components of a healthy and nutritionally adequate diet
- ***Focus on disease prevention*** rather than disease *treatment*
- Inform Federal food, nutrition, and health policies and programs

Key Elements of Healthy Eating Patterns:

Key Recommendations

 **Consume a healthy eating pattern that accounts for all foods and beverages *within an appropriate calorie level*.**

 **A healthy eating pattern *includes*:**

- A variety of vegetables from all of the subgroups—dark green, red and orange, legumes (beans and peas), starchy, and other
- Fruits, especially whole fruits
- Grains, at least half of which are whole grains
- Fat-free or low-fat dairy, including milk, yogurt, cheese, and/or fortified soy beverages
- A variety of protein foods, including seafood, lean meats and poultry, eggs, legumes (beans and peas), and nuts, seeds, and soy products
- Oils

 **A healthy eating pattern *limits*:**

- Saturated fats and *trans* fats, added sugars, and sodium

Key Elements of Healthy Eating Patterns:

Key Recommendations (cont.)

Key Recommendations that are quantitative are provided for several components of the diet that should be limited. These components are of particular public health concern in the United States, and the specified limits can help individuals achieve healthy eating patterns within calorie limits:

- ❦ ***Consume less than 10 percent of calories per day from added sugars***
- ❦ ***Consume less than 10 percent of calories per day from saturated fats***
- ❦ ***Consume less than 2,300 milligrams (mg) per day of sodium***
- ❦ If alcohol is consumed, it should be consumed in moderation—up to one drink per day for women and up to two drinks per day for men—and only by adults of legal drinking age
- ❦ Meet the *Physical Activity Guidelines for Americans*

Principles of Healthy Eating Patterns

- An eating pattern represents the totality of all foods and beverages consumed
 - It is more than the sum of its parts; the totality of what individuals habitually eat and drink act synergistically in relation to health.
- Nutritional needs should be met primarily from foods
 - Individuals should aim to meet their nutrient needs through healthy eating patterns that include foods in nutrient-dense forms.
- Healthy eating patterns are adaptable
 - Any eating pattern can be tailored to the individual's socio-cultural and personal preferences.

Inside Healthy Eating Patterns:

Food Groups

“Eating an appropriate mix of foods from the food groups and subgroups—within an appropriate calorie level—is important to promote health.”

- Each food group and subgroup provides an array of nutrients.
- Recommended amounts reflect eating patterns associated with positive health outcomes.
- Foods from all of the food groups should be eaten in nutrient-dense forms.

Inside Healthy Eating Patterns:

Food Groups — Examples of Content



Vegetables

- Healthy eating patterns include a variety of vegetables from all five vegetable subgroups— **dark green**, **red and orange**, legumes (beans and peas), starchy, and other.
- Include all fresh, frozen, canned, and dried options in cooked or raw forms, including vegetable juices.
- Vegetables should be consumed in a nutrient-dense form, with ***limited additions such as salt, butter, or creamy sauces.***



Dairy

- Healthy eating patterns include fat-free and low-fat (1%) dairy, including milk, yogurt, cheese, or fortified soy beverages (soymilk).
- Fat-free or low-fat milk and yogurt, in comparison to cheese, contain less saturated fats and sodium and more potassium, vitamin A, and vitamin D.

Inside Healthy Eating Patterns:

Other Components

“In addition to the food groups, it is important to consider other food components when making food and beverage choices.”

- Added sugars
- Saturated fats
- *Trans* fats
- Cholesterol
- Sodium
- Alcohol
- Caffeine

Inside Healthy Eating Patterns:

Other Components — Examples of Content



Cholesterol

- Individuals should eat as little dietary cholesterol as possible while consuming a healthy eating pattern.
- The Healthy U.S.-Style Eating Pattern contains ***approximately 100 to 300 mg of cholesterol*** across the 12 calorie levels.



Caffeine

- Much of the available evidence on caffeine focuses on coffee intake.
- ***Three to five 8-oz cups of coffee per day*** can be incorporated into healthy eating patterns.
- Individuals who do not consume caffeinated beverages are not encouraged to incorporate them into their eating pattern.
- In addition, caffeinated beverages may contain added calories from ***cream, whole or 2% milk, creamer, and added sugars, which should be limited.***

Shifts to Align with Healthy Eating Patterns: Examples

- **Increasing vegetables** in mixed dishes while **decreasing the amounts of refined grains or meats high in saturated fat and/or sodium**.
- **Incorporating seafood** in meals twice per week in place of meat, poultry, or eggs.
- **Using vegetable oil** in place of solid fats when cooking, and using oil-based dressings and spreads on foods instead of those made from solid fats.
- Choosing beverages with **no added sugars**, such as water.
- Using the Nutrition Facts label to compare **sodium** content of foods and choosing the product with less sodium.

Nutrient Dense Foods: Foods with little calories but lots of nutrients!



Empty Calories

Foods with lots of calories from fats and sugars and little or no nutrients.



Nutrition Assessment - Anthropometry

- Height
- Weight
- BMI
- Waist Circumference
- % Body Fat

Anthropometry: BMI, Waist Circumference and Disease Risk

TABLE
9.3

Classification of Overweight and Obesity by BMI, Waist Circumference, and Associated Disease Risks

	BMI (kg/m ²)	Obesity Class	Disease Risk* Relative to Normal Weight and Waist Circumference	
			Men 102 cm (40 in.) or less Women 88 cm (35 in.) or less	Men > 102 cm (40 in.) Women > 88 cm (35 in.)
Underweight	< 18.5		—	—
Normal	18.5–24.9		—	—
Overweight	25–29.9		Increased	High
Obesity	30–34.9	I	High	Very high
	35–39.9	II	Very high	Very high
Extreme Obesity	40.0 [†]	III	Extremely high	Extremely high

* Disease risk for type 2 diabetes, hypertension, and cardiovascular disease.

[†] Increased waist circumference can also be a marker for increased risk even in persons of normal weight.

Source: Data from National Heart, Lung and Blood Institute, 2008.

Anthropometry: % Body Fat Norms

American Council on Exercise

ACE Body Fat % Chart		
Description	Women	Men
Essential fat	10-13%	2-5%
Athletes	14–20%	6-13%
Fitness	21–24%	14–17%
Average	25–31%	18–24%
Obese	32%+	25%+

Energy Balance- simple model

Calories Consumed



Calories Burned



**Energy
Balance**

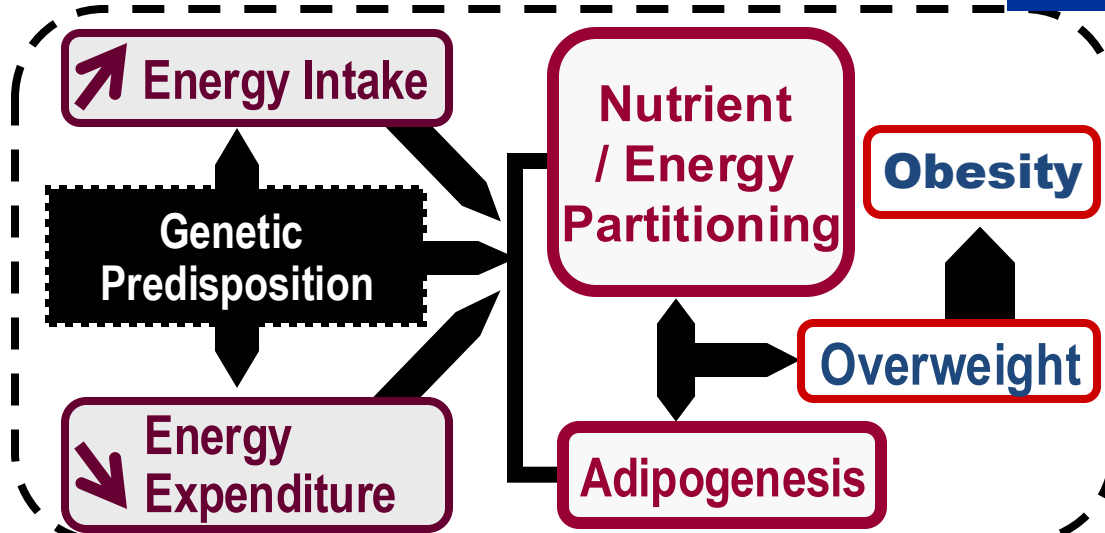
Social Environment

- Eating on the run
- Eating as recreation
- Pressure to be sedentary
- Fewer meals at home
- Powerful and constant advertising
- Society of spectators instead of participant

Physical Environment

- Pressure to consume
- Building design
- Absence of sidewalks
- Population density
- Urban sprawl
- Pollutants
- Automobile dependence
- Certain medications

- Calorie-dense foods
- Maternal-fetal nutrition
- More sedentarism
- Lactation
- High fat diets
- Larger portions



- Corn fructose syrup
- Less physical activity
- Smoking cessation

Behavior

Thermogenesis

n-6/n-3 PUFAs

Lipid ox

Viruses

RMR

Genetic hypotheses

CNS regulators of appetite: NPY, α -MSH, CART, Orexins, Agouti, MC4R, MCH, AGRP, etc.

Regulators of adipogenesis: RAR, RXR, PPAR γ , C/EBP, SREBP-1c, PGC-1, etc.

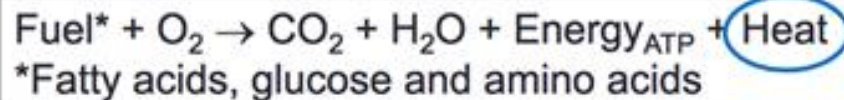
Epigenetics

Biology

Peripheral regulators of appetite: PYY, insulin, leptin, ghrelin, CCK, GLP-1, etc.

Energy Expenditure

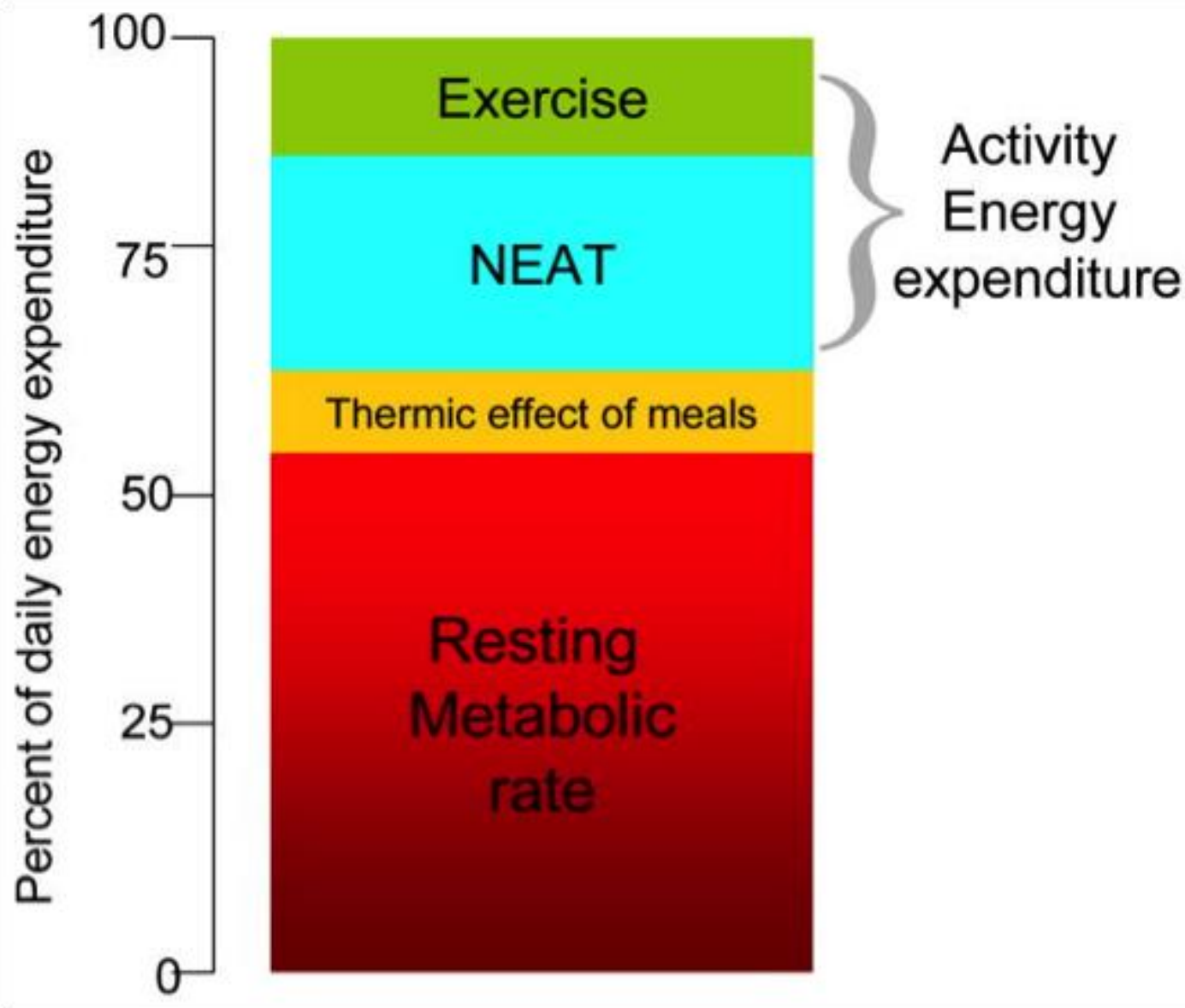
- Energy expenditure is divided into three categories:
 - Resting metabolic rate (RMR) aka basal metabolic rate (BMR)
 - Physical activity (MET)
 - Thermic Effect of Food (TEF)
- RMR is determined by measuring the heat given off as a byproduct of fuel metabolism



Total Energy Expenditure (TEE)



Three primary components of energy expenditure



Non Exercise Activity Thermogenesis (NEAT)

This is the amount of calories your body uses for everything that's not eating, exercise, BMR or sleeping. Based on activity & can make up 10-40% of your total caloric burn - Based on activity level



Walking



Fidgeting



Standing



Doing the dishes
or cooking

How To Increase Your Daily NEAT



Walk
More



Get a
Pedometer



Use a
Standing
Desk



Clean Up
Around The
House More

Assessing Nutrient Intake

- Assess energy requirements via Harris Benedict Equation.
- Compare current caloric intake with calculated requirements.

Estimation of Adult Resting Energy Requirements (REE)

HOW TO CALCULATE YOUR CALORIES

Basal Metabolic Rate (BMR) is the number of **calories** you would burn with NO activity.



MEN

$$\text{BMR} = 66 + (6.23 \times \text{weight in lbs}) + (12.7 \times \text{height in inches}) - (6.8 \times \text{age})$$



WOMEN

$$\text{BMR} = 655 + (4.35 \times \text{weight in lbs}) + (4.7 \times \text{height in inches}) - (4.7 \times \text{age})$$

(TIP: use **Lean Body weight** (% body fat x weight in lbs) if possible)

YOUR TARGET DAILY CALORIE NEEDS

- 1 Little or no exercise: BMR x 1.2
- 2 Light Exercise/sports 1-3 days/week: BMR x 1.375
- 3 Medium Exercise/sports 3-5 days/week: BMR x 1.55
- 4 Hard Exercise/sports 6-7 days a week: BMR x 1.725
- 5 Intense exercise/sports, physical job or twice/day training): BMR x 1.9

Macronutrients – Energy Producing

- **Carbohydrates (CHO)**

- Provides energy

- **Proteins**

- Builds/repairs body tissue

- **Fats (lipids)**

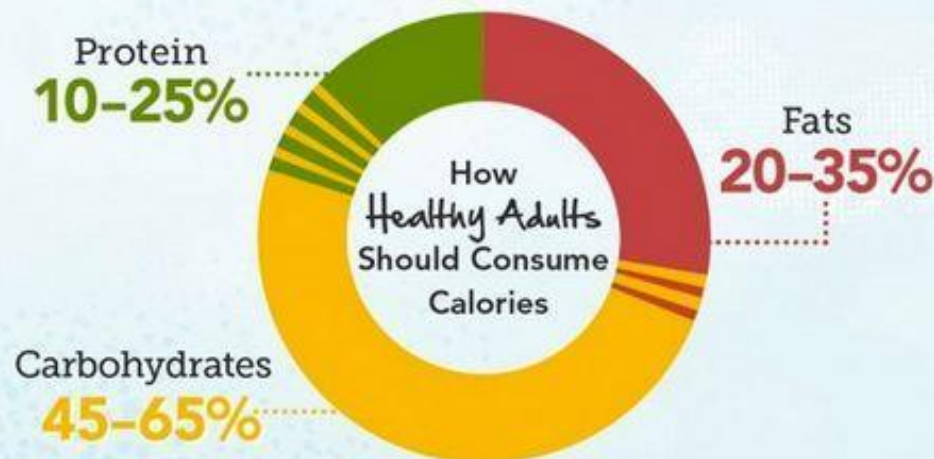
- Provides insulation/energy, protects organs

- Important for growth and development
- Act to keep body functioning normally
- Provide energy (*kilocalorie*)

- Kcal: measure of the amount of heat needed to raise the temperature of 1000 grams (1 liter) of water to 1 degree C



THIS IS HOW MANY CALORIES ARE IN 1 gram OF EACH MACRONUTRIENT:



CARBS

WHY WE NEED THEM

Carbs are the body's main source of fuel and are easily used by the body for energy.

HOW THEY'RE USED

Carbs break down into glucose, which is used for energy. They're also stored in muscles and liver for later use and are important for the central nervous system, kidneys, brain, muscles, and intestinal health.

WHERE TO FIND THEM

Grains, fruits, milk and yogurt.



PROTEIN

Protein is essential for growth, tissue repair, immune function, preserving lean muscle, and producing essential hormones and enzymes.

The body uses protein for energy when carbs aren't available.

Meat, poultry, fish, cheese, milk, nuts, and legumes.



FATS

Fats are essential in cell, nerve tissue, and hormone production. Fats are also essential for absorbing fat-soluble vitamins like vitamins A, D, E, K, and carotenoids.

Fats are the most concentrated source of energy. If fats consumed aren't burned as energy or used to build body tissues, they're stored in the body's fat cells for later use.

Unsaturated fats such as olive oil, avocado, nuts, fatty fish (like salmon) and canola oil.



Caloric Breakdown

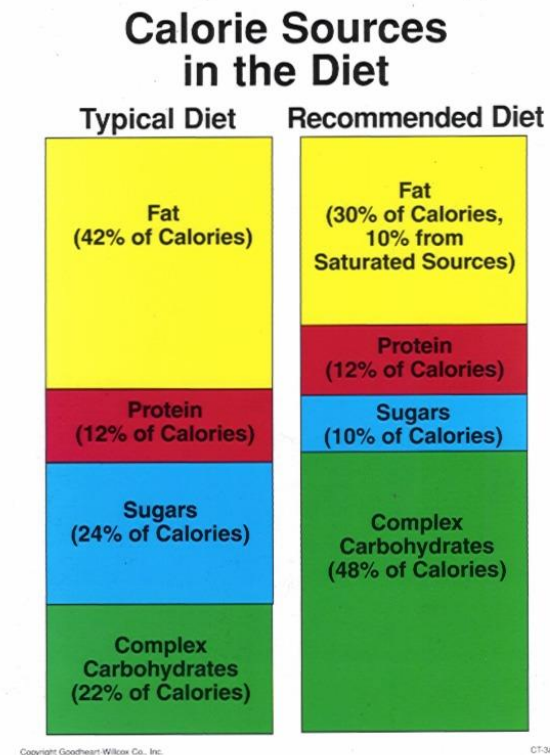
Carbohydrates: 55-60%

Fat: No more than 30%

Protein: 10-15 %

The Average American:

- Consumes too many calories from saturated fat, sugar, & sodium.
- Doesn't consume enough fiber.



Wait ... What About Water?
Yes ... it's a Macronutrient!

Steps to Promote Optimal Weight and Nutrition

- Are you hydrated?
 - If your urine is a pale yellow ... then yes!
 - Daily water intake (rough estimate):

How much water should you drink?

your body weight (lbs.)

2

=



fluid oz. of
water daily

NEW LABEL / WHAT'S DIFFERENT

Servings:
larger,
bolder type

Nutrition Facts

8 servings per container

Serving size 2/3 cup (55g)

Amount per serving

Calories 230

% Daily Value*

Total Fat 8g 10%

Saturated Fat 1g 5%

Trans Fat 0g

Cholesterol 0mg 0%

Sodium 160mg 7%

Total Carbohydrate 37g 13%

Dietary Fiber 4g 14%

Total Sugars 12g

Includes 10g Added Sugars 20%

Protein 3g

Vitamin D 2mcg 10%

Calcium 260mg 20%

Iron 8mg 45%

Potassium 235mg 6%

* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

Serving sizes
updated

Calories:
larger type

Updated
daily
values

Actual
amounts
declared

New
footnote

New:
added sugars

Change
in nutrients
required

Unsaturated Fats/Saturated Fats

Good Fat

- **Unsaturated Fats**
 - **Monounsaturated fat** [MUFA] - (*healthiest*)
olive, sesame, corn)
 - **Polyunsaturated fat** [PUFA] - (*spoil quicker*)
corn oils, sunflower oils
- *Liquid* at room temperature

Bad Fat

- **Saturated Fats**
 - The “fat molecule” is *fully saturated* with hydrogen.
 - There are no double bonds between the carbon atoms.
 - Examples: animal products, cheese
 - ***Solid*** at room temperature
- **Trans-fats** – *Partially hydrogenated oils (PHOs)*
 - Process: they heat unsaturated oil and add hydrogen to make it more saturated.
 - ***Worst kind of fat***, limit to a few grams a day!
 - **Banned in US in 2015**

Carbohydrates - CHO (Starch, Sugar, Fiber)

SIMPLE CHO

- **Monosaccharides** (one unit sugars)
 - Glucose, fructose, galactose
 - Absorbed directly, no breakdown
- **Disaccharides** (formed when 2 monosaccharides combine)
 - Lactose: glucose and galactose (found in milk and dairy)
 - Sucrose: glucose and fructose (table sugar, also naturally in some fruits and other foods)



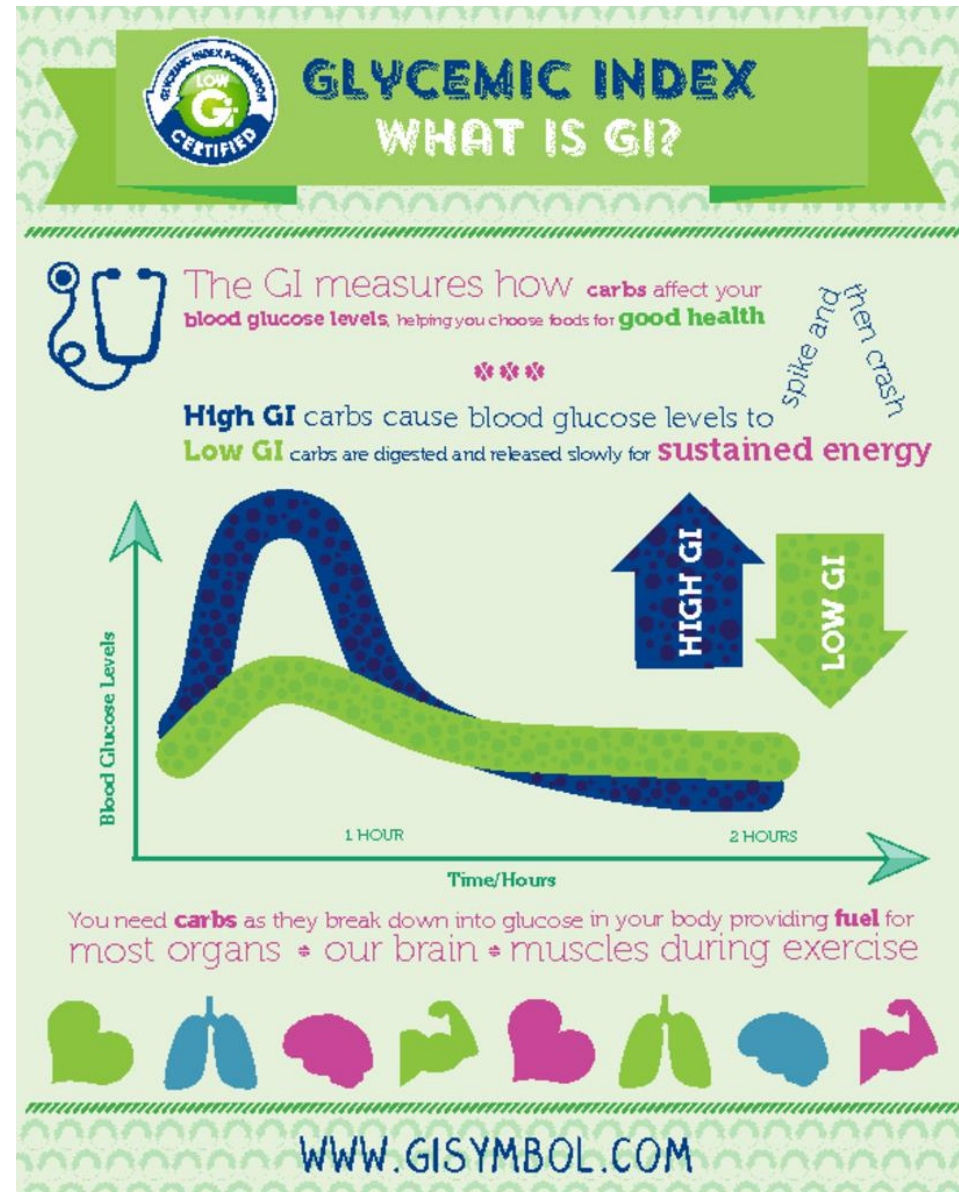
COMPLEX CHO

- **Polysaccharides** - many units of sugar
 - Both Fiber and Starch
 - *Plant foods, grains, potatoes, and legumes contain starches*
 - All CHO except for fiber are broken down to simple sugars
 - **Fiber:**
 - *Not entirely broken down*
 - *Some stays in digestive system*
 - *Health benefits*



Glycemic Index

- CHO_s → glucose + other monosaccharides
 - → increases blood glucose + supplies energy
- **Simple CHO** - small molecules
 - Raise blood glucose rapidly
 - High glycemic index
- **Complex CHO** - larger molecules
 - Raise blood sugar less rapid
 - Low glycemic index



2020-2025 Dietary Guidelines

- **Saturated fat:** **<10% of diet**
- **Sodium:** **< 2300 mg/day**
- **Sugar:** <100 cal/day female [<150 male] → **<10% diet**
- **Cholesterol:** does not have quantitative limit but should eat as little as possible [**<300 cal/day**]
- **Caffeine:** limits 400 mg daily (**3-4 cups**)



2015-2020 Dietary Guidelines for Americans | Added Sugars

Consume less than 10% of calories per day from added sugars.

If you eat 2000 calories per day, the added sugars you eat should be less than 200 calories, or 50 g of sugar. This is about the amount of added sugar in a 12-oz bottle of soda.

1 tsp of sugar = 4 g of sugar
(16 calories)



Biggest sources of added sugars

Regular soda, sugared beverages,
most sports drinks

Cakes and cookies

Candies

Ice cream

Pies



Hidden sources of added sugars

Flavorings added to coffee,
tea, or other beverages

Cereals

Flavored yogurts

Snack bars



K. BUCHER

2015-2020 Dietary Guidelines for Americans | Salt

Limit sodium intake to 2300 mg per day.

For those with prehypertension and hypertension, limit sodium intake to 1500 mg per day because reducing sodium intake can lower blood pressure.



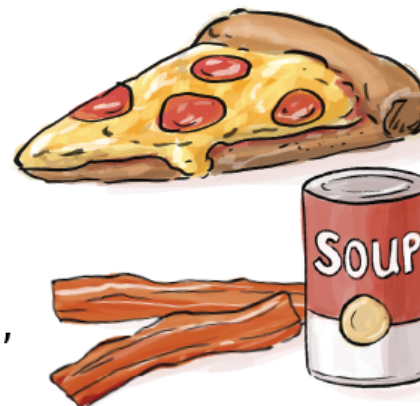
$\frac{1}{2}$ tsp of salt =
1200 mg of sodium

Biggest sources of sodium

Processed and ready-made foods

Restaurant and fast foods, including pizza, hamburgers, and sandwiches

Store-bought and processed foods, including deli meats, bacon, ham, sausage, frozen items, canned items, soups, sauces, and dressings

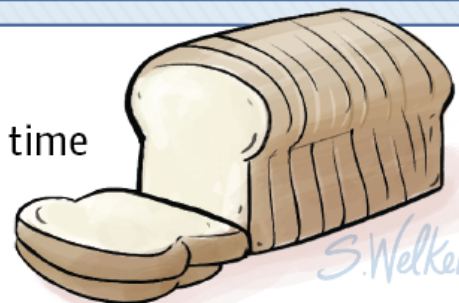


Hidden sources of sodium

Shelf-stable products designed to last a long time

Bakery products, including sliced bread

Cheese



WHAT'S **A SERVING?**

FRUITS

4 servings per day

ONE MEDIUM FRUIT

 = About the size
of your fist

FRESH, FROZEN OR CANNED

 = 1/2 CUP

DRIED

 = 1/4 CUP

FRUIT JUICE

 = 1/4 CUP

VEGETABLES

5 servings per day

RAW LEAFY VEGETABLE

 = 1 CUP

FRESH, FROZEN OR CANNED

 = 1/2 CUP

VEGETABLE JUICE

 = 1/2 CUP

*based on a 2,000 calorie eating pattern

The good news is eating the right amount of fruits and vegetables doesn't have to be complicated. Here are some examples of about one serving:

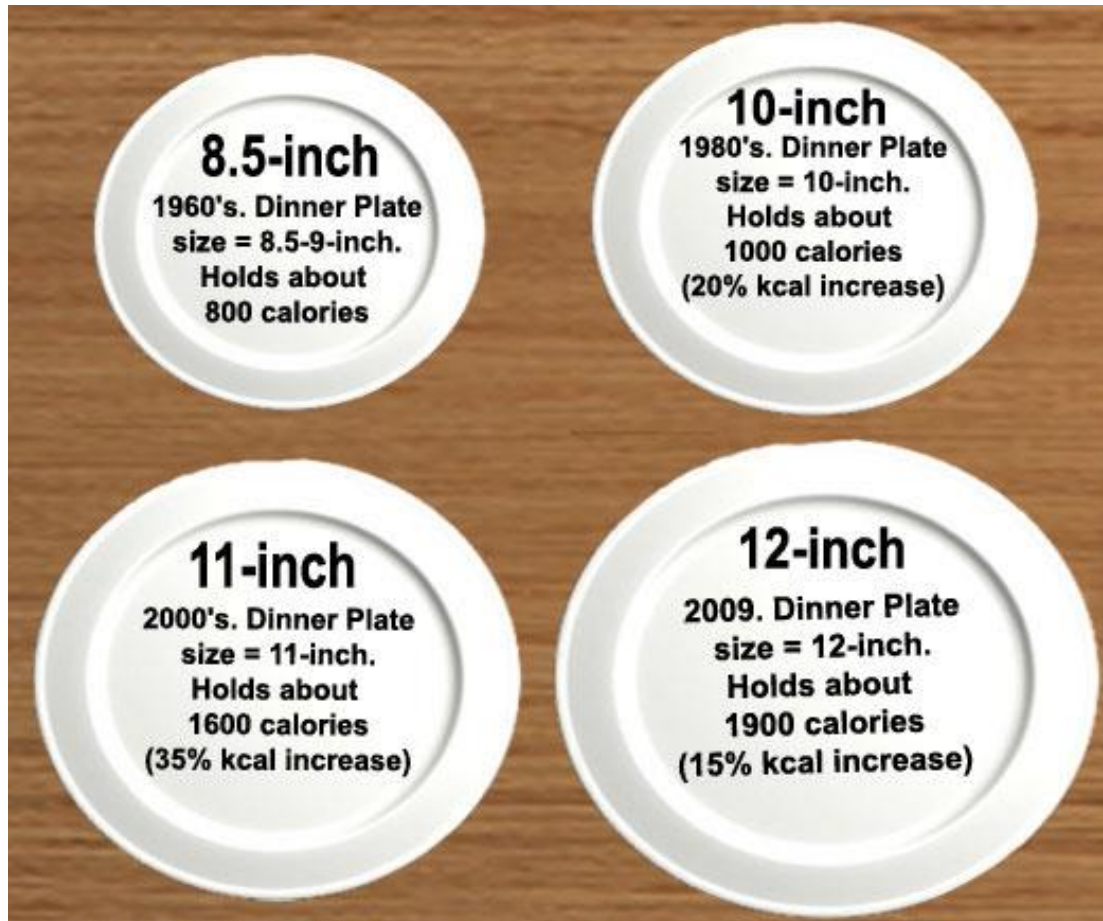
FRUITS

Apple, pear, orange, peach or nectarine: 1 medium
Avocado: Half of a medium
Banana: 1 small (about 6" long)
Grapefruit: Half of a medium (4" across)
Grape: 16
Kiwifruit: 1 medium
Mango: Half of a medium
Melon: Half-inch thick wedge of sliced watermelon, honeydew, cantaloupe
Pineapple: 1/4 of a medium
Strawberry: 4 large

VEGETABLES

Bell pepper: Half of a large
Broccoli or cauliflower: 5 to 8 florets
Carrot: 6 baby or 1 whole medium (6 to 7" long)
Corn: 1 small ear (6" long) or half of a large ear (8 to 9" long)
Leafy vegetable: 1 cup raw or 1/2 cup cooked (lettuce, kale, spinach, greens)
Potato: Half of a medium (2 1/2 to 3" across)
Squash, yellow: Half of a small
Sweet potato: Half of a large (2 1/4" across)
Zucchini: Half of a large (7 to 8" long)

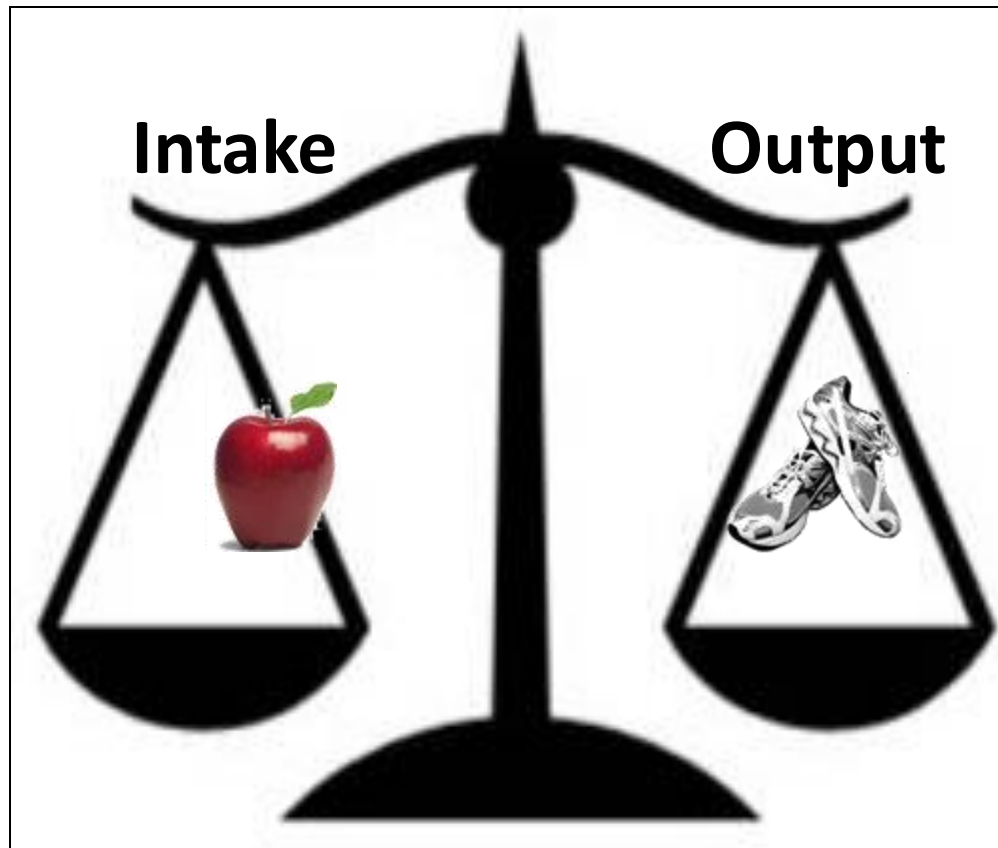
Plate size history



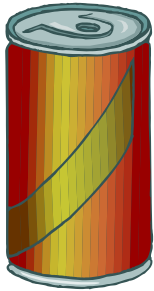
Stay within your daily calorie needs

**100 *extra*
calories per day**

**10 pound
weight gain per year**



Examples of 100 calories



- $\frac{2}{3}$ can of a regular soft drink



- 2 tablespoons maple syrup
- 1 tablespoon butter or margarine
- 2 tablespoons jelly or jam



10 large jelly beans (1 oz.)

2. Focus on variety, nutrient density, and amount.

- o To meet nutrient needs within calorie limits, *choose a variety of nutrient-dense foods* across and within all food groups in recommended amounts.
- o Nutrient dense foods provide vitamins, minerals and other beneficial substances with relatively few calories.

Which is more Nutrient Dense?

Spinach



OR



Candy

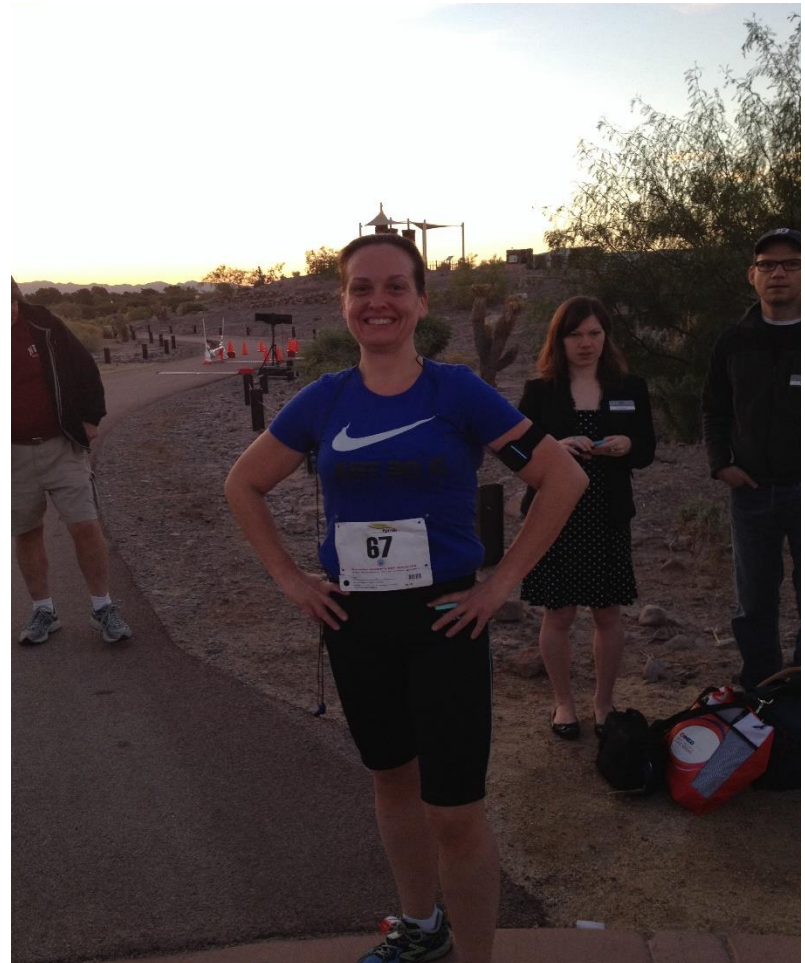
Summary Slide

- **Saturated fat:** <10% of diet [beginning at age 2]
- **Sodium:** < 2300 mg/day [Ages 14 and older]
 - Or even less if younger than age 14]
- **Sugar:** <100 cal/day female [<150 male] → <10% diet [Ages 2 and older]
 - AVOID for infants and toddlers
- **Alcohol [if consumed]:** Limiting to 2 drinks or less a day for men and 1 drink or less a day for women.
- **BMI, Waist Circumference and Disease Risk** → See Table

Core References

- https://www.dietaryguidelines.gov/sites/default/files/2020-12/Dietary_Guidelines_for_Americans_2020-2025.pdf
- <https://health.gov/our-work/food-nutrition/2015-2020-dietary-guidelines/guidelines/>
- <https://www.choosemyplate.gov/eathealthy/dietary-guidelines>
- <https://www.nhlbi.nih.gov/science/obesity-nutrition-and-physical-activity>

Thank You! Any Questions?



Lecture Feedback Form:

<https://www.surveymonkey.com/r/NYITCOMLectureFeedback>

Reference Slides

Slides 65-85

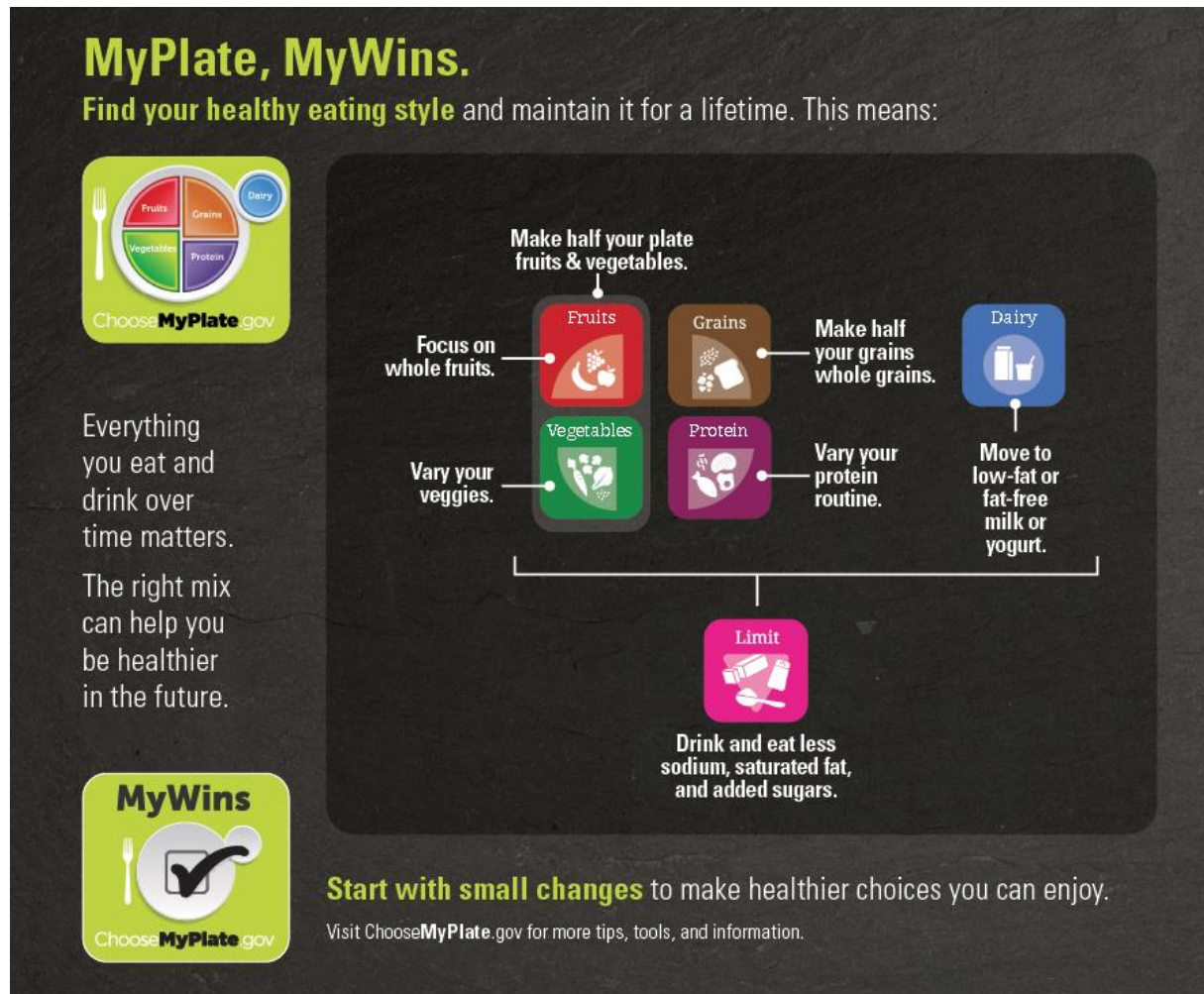
*Additional Slides
For Your Information*

<https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials/food-sources-select-nutrients>

<https://www.dietaryguidelines.gov/resources/consumer-resources>

Implementing the Guidelines Through MyPlate

(Figure 3-2)



HEALTHY EATING PLATE

Use healthy oils (like olive and canola oil) for cooking, on salad, and at the table. Limit butter. Avoid trans fat.



The more veggies – and the greater the variety – the better. Potatoes and French fries don't count.

Eat plenty of fruits of all colors.



STAY ACTIVE!

© Harvard University



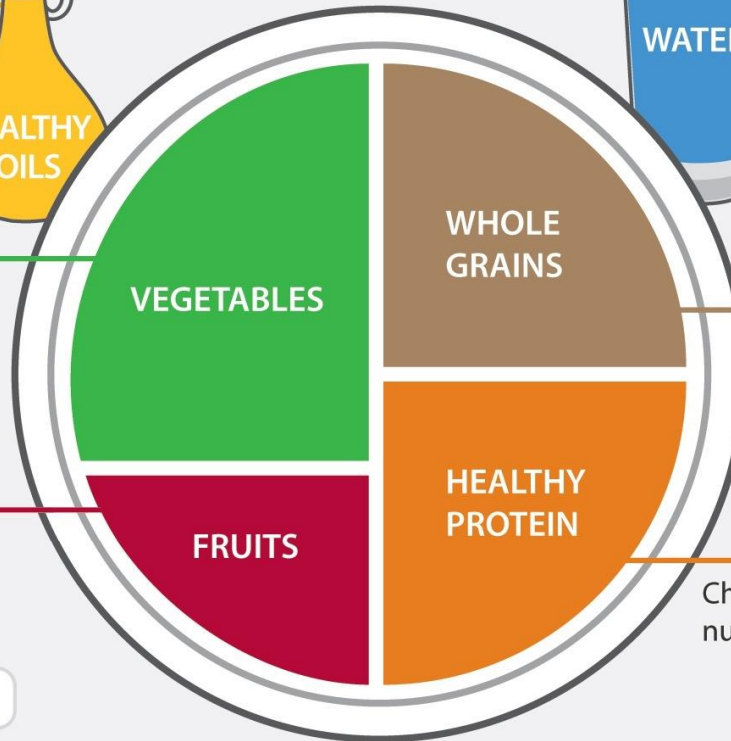
Harvard School of Public Health
The Nutrition Source
www.hsph.harvard.edu/nutritionsource



Drink water, tea, or coffee (with little or no sugar). Limit milk/dairy (1-2 servings/day) and juice (1 small glass/day). Avoid sugary drinks.

Eat a variety of whole grains (like whole-wheat bread, whole-grain pasta, and brown rice). Limit refined grains (like white rice and white bread).

Choose fish, poultry, beans, and nuts; limit red meat and cheese; avoid bacon, cold cuts, and other processed meats.



Harvard Medical School
Harvard Health Publications
www.health.harvard.edu



GRAINS

Make half your grains whole

Eat at least 3 oz. of whole-grain cereals, breads, crackers, rice, or pasta every day

1 oz. is about 1 slice of bread, about 1 cup of breakfast cereal, or 1/2 cup of cooked rice, cereal, or pasta

VEGETABLES

Vary your veggies

Eat more dark-green veggies like broccoli, spinach, and other dark leafy greens

Eat more orange vegetables like carrots and sweetpotatoes

Eat more dry beans and peas like pinto beans, kidney beans, and lentils

FRUITS

Focus on fruits

Eat a variety of fruit

Choose fresh, frozen, canned, or dried fruit

Go easy on fruit juices

MILK

Get your calcium-rich foods

Go low-fat or fat-free when you choose milk, yogurt, and other milk products

If you don't or can't consume milk, choose lactose-free products or other calcium sources such as fortified foods and beverages

MEAT & BEANS

Go lean with protein

Choose low-fat or lean meats and poultry

Bake it, broil it, or grill it

Vary your protein routine — choose more fish, beans, peas, nuts, and seeds

For a 2,000-calorie diet, you need the amounts below from each food group. To find the amounts that are right for you, go to MyPyramid.gov.

Eat 6 oz. every day

Eat 2 1/2 cups every day

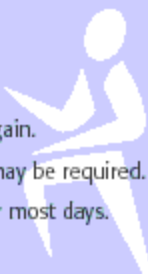
Eat 2 cups every day

Get 3 cups every day;
for kids aged 2 to 8, it's 2

Eat 5 1/2 oz. every day

Find your balance between food and physical activity

- Be sure to stay within your daily calorie needs.
- Be physically active for at least 30 minutes most days of the week.
- About 60 minutes a day of physical activity may be needed to prevent weight gain.
- For sustaining weight loss, at least 60 to 90 minutes a day of physical activity may be required.
- Children and teenagers should be physically active for 60 minutes every day, or most days.



Know the limits on fats, sugars, and salt (sodium)

- Make most of your fat sources from fish, nuts, and vegetable oils.
- Limit solid fats like butter, stick margarine, shortening, and lard, as well as foods that contain these.
- Check the Nutrition Facts label to keep saturated fats, *trans* fats, and sodium low.
- Choose food and beverages low in added sugars. Added sugars contribute calories with few, if any, nutrients.



U.S. Department of Agriculture
Center for Nutrition Policy and Promotion
April 2005
CNPP-15



USDA is an equal opportunity provider and employer.

Daily Values (From FDA)

- A portion, or percentage of our total calorie needs in a day
- Based on a 2,000 Calorie Diet per day
- Derived from RDAs and other sources
- **Not everyone needs 2,000 kcals per day!**
- According to the Dietary Guidelines for Americans, the recommended macronutrient intake ranges are:
 - Carbs: 45-65%
 - Fat: 20-35%
 - Protein: 10-30%

* Percent Daily Values are based on a 2,000 calorie diet. Your Daily Values may be higher or lower depending on your calorie needs:

		Calories:	2,000	2,500
Total Fat	Less than		65g	80g
Sat Fat	Less than		20g	25g
Cholesterol	Less than		300mg	300mg
Sodium	Less than		2,400mg	2,400mg
Total Carbohydrate			300g	375g
Dietary Fiber			25g	30g

Assessment of Nutrient Intake (Food Labels)

- **Dietary Reference Intakes (DRIs)**
 - Reference values that are quantitative of nutrient intakes to be used for planning and assessing diets for healthy people.
- **Recommended Dietary Allowance(RDAs)**
 - Recommended nutrient intakes that meet the needs of essentially all people of similar age and gender.

THE ROLES OF Micronutrients

The body uses dozens of different beneficial micronutrient chemicals every hour of every day to keep us energized, produce enzymes and hormones, and prevent deficiencies.

Low levels of specific micronutrients can result in various problems like:

- *mental impairment*
- *poor digestion*
- *thyroid problems*
- *bone loss*



THE JOB OF Micronutrients

INCLUDES

- Producing digestive enzymes
- Helping keep a strong metabolism
- Breaking down carbs, fats and proteins into usable energy
- Helping with hormone production
- Slowing oxidation damage or signs of aging caused by free radicals
- Protecting the brain
- Aiding in bone mineralization
- Synthesizing DNA
- Allowing cells to rejuvenate
- Allowing muscles to move and helping with tissue repair
- Facilitating growth

MICRO Nutrients

VS

MACRO Nutrients

**THE
CHEMICALS**
found in trace
amounts in foods



More than one
present in
most foods



The way we classify
**CALORIES IN
FOODS**



More than one
present in
most foods



There are 3 main groups

**VITAMINS
MINERALS
ANTIOXIDANTS**

There are 3 main groups

**FATS
CARBS
PROTEINS**

LESS FAMILIAR

because we can't always
easily pinpoint which
ones are in which foods

MORE FAMILIAR

because they're how
we classify foods

Comprise
"small picture"
food elements because
we need them in
smaller quantities

Comprise
"BIG PICTURE"
food elements because
we need them in
larger quantities

Micronutrients and Macronutrients

Figure 1.1 **Micronutrients versus Macronutrients** Vitamins and minerals are micronutrients, whereas water, carbohydrates, proteins, and lipids are macronutrients.

We need only small amounts of the micronutrients.

Micronutrients

- Vitamins
- Minerals

Macronutrients

- Water
- Carbohydrates
- Proteins
- Lipids (fats)

We need larger amounts of the macronutrients.

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Table. Simple classification of dietary constituents

Constituent	Use
Water	To provide body fluid and to help regulate body temperature
Carbohydrates	As fuel for energy for body heat and work
Fats	As fuel for energy and essential fatty acids
Proteins	For growth and repair
Minerals	For developing body tissues and for metabolic processes and protection
Vitamins	For metabolic processes and protection
Indigestible and unabsorbable particles, including fibre	To form a vehicle for other nutrients, add bulk to the diet, provide a habitat for bacterial flora and assist proper elimination of refuse

Added Sugars: Intakes and Limit

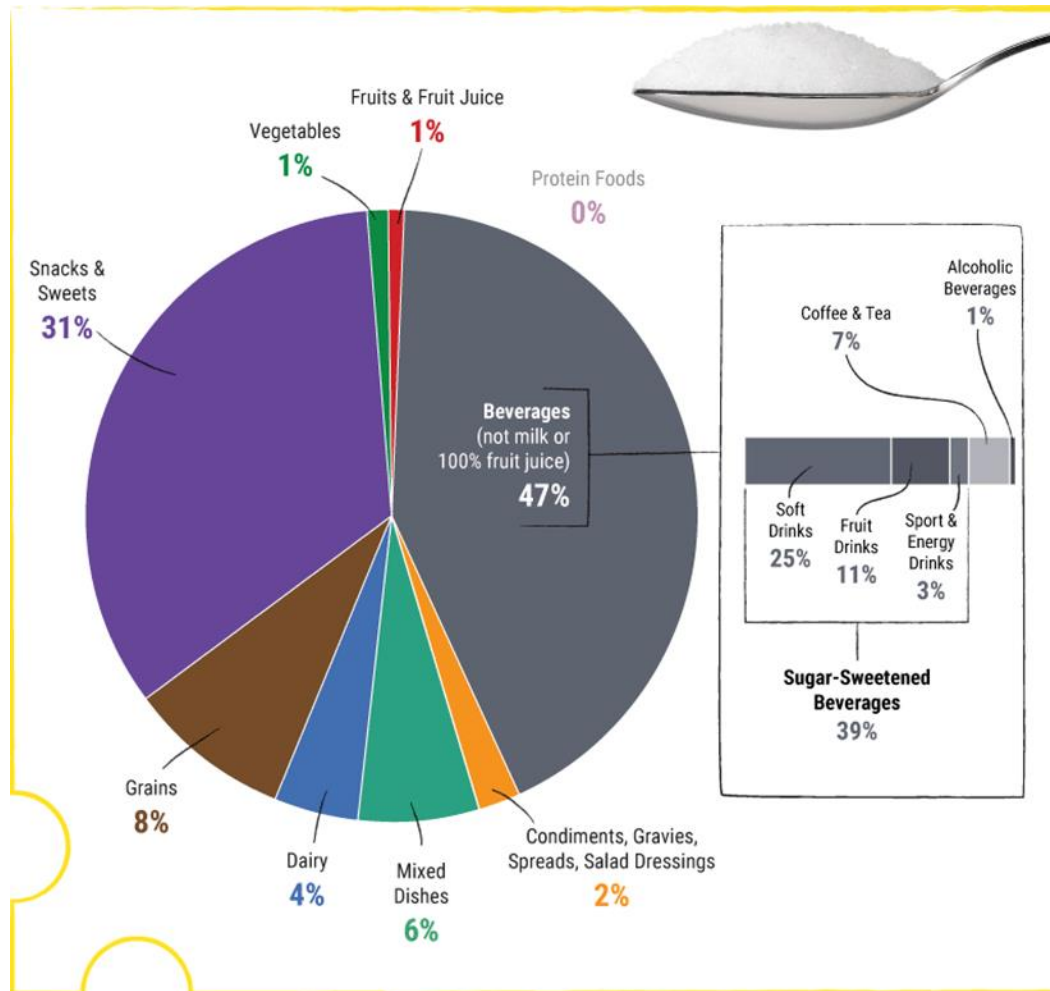
Average Intakes as a Percent of Calories per Day by Age-Sex Group, in Comparison to the Dietary Guidelines Maximum Limit of Less than 10 Percent of Calories (Figure 2-9)



- **Note:**
The maximum amount of added sugars allowable in a Healthy U.S.-

Food Sources of Added Sugars

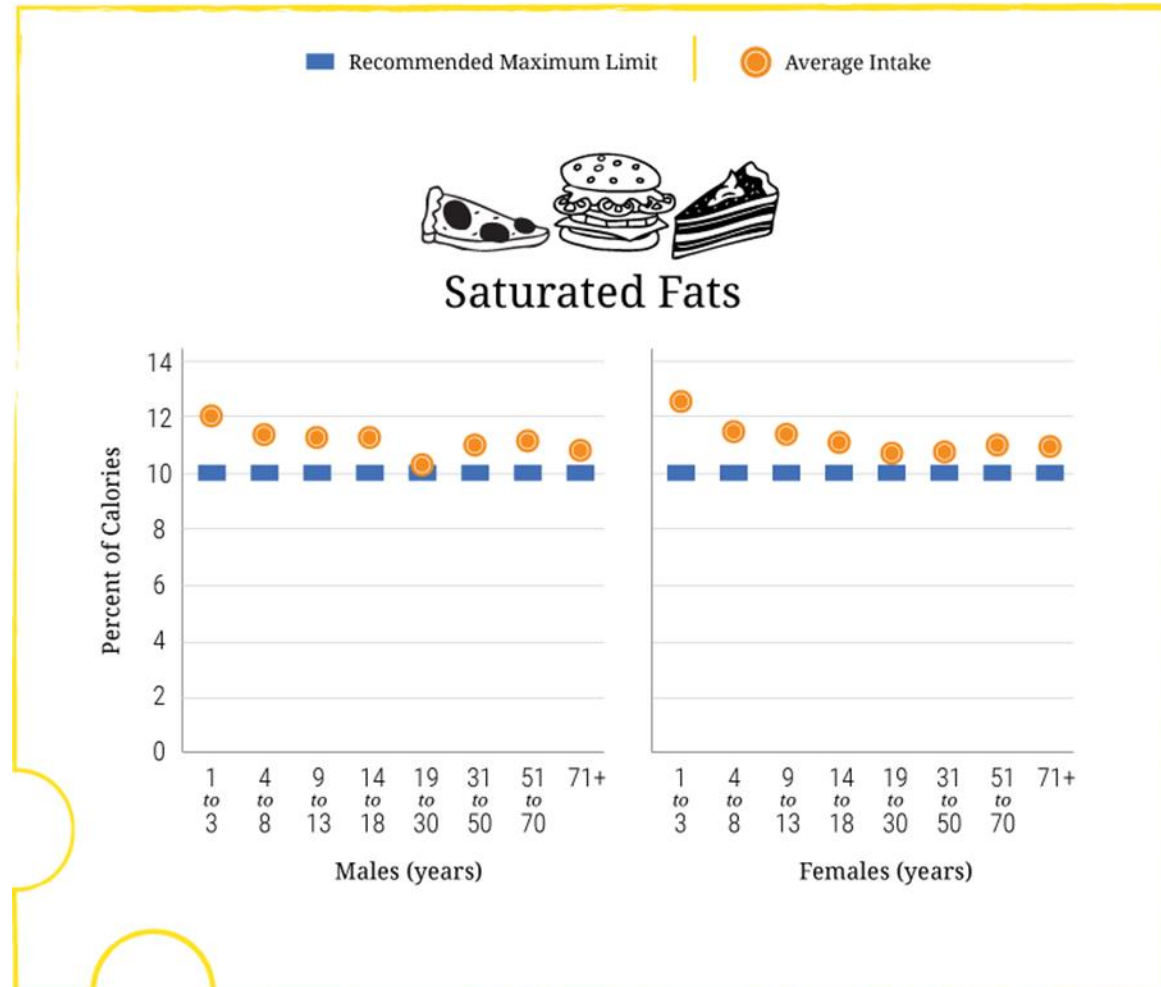
Food Category Sources of Added Sugars in the U.S. Population Ages 2 Years and Older (Figure 2-10)



- **Data Source:** What We Eat in America (WWEIA) Food Category analysis

Saturated Fats: Intakes and Limit

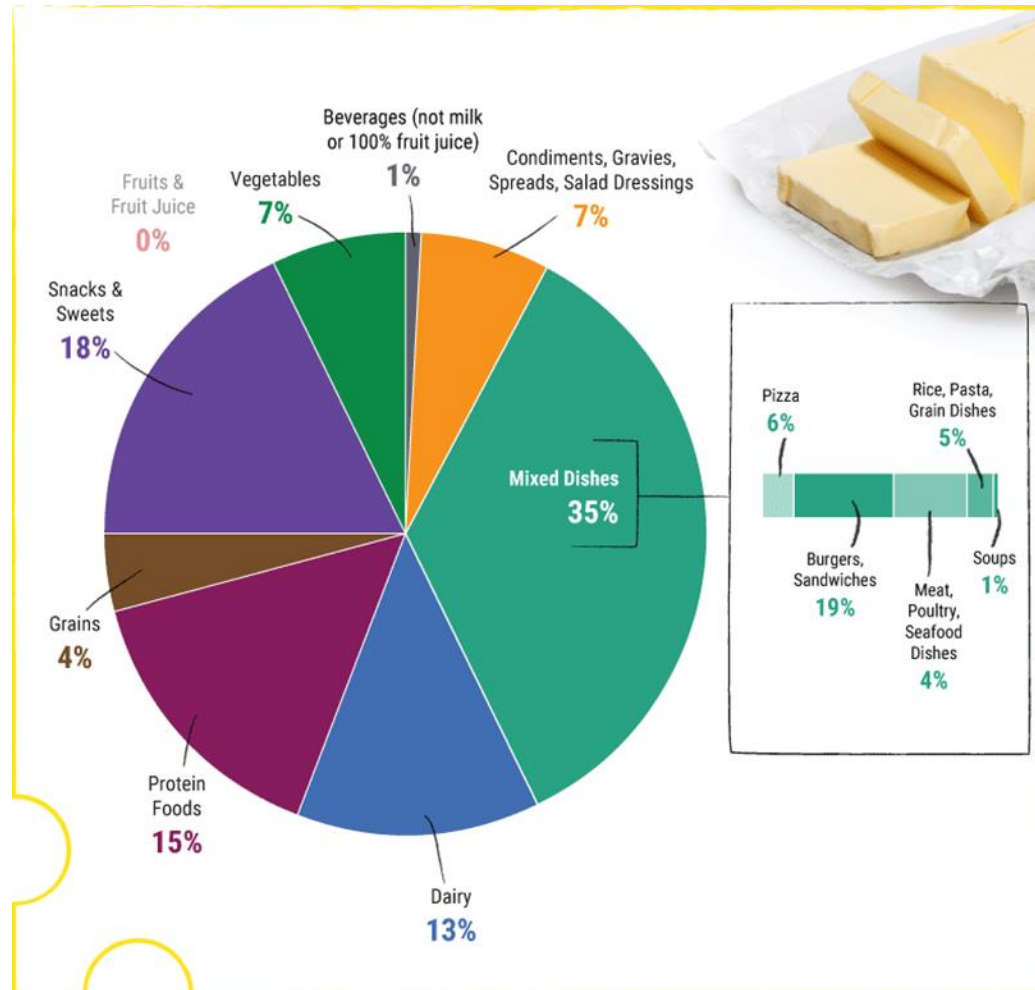
Average Intakes as a Percent of Calories per Day by Age-Sex Group, in Comparison to the Dietary Guidelines Maximum Limit of Less Than 10 Percent of Calories (Figure 2-11)



- **Data Source:** What We Eat in America, NHANES 2007-2010 for average

Food Sources of Saturated Fats

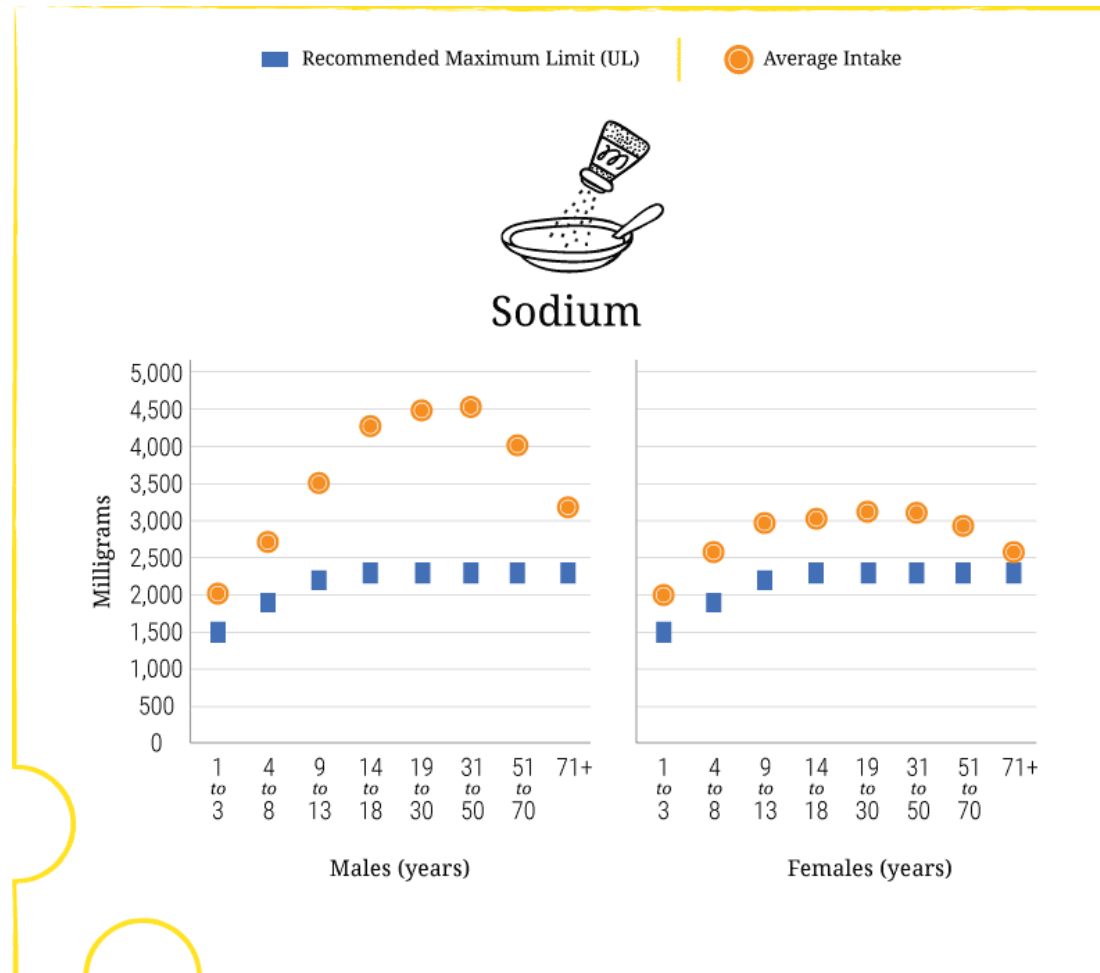
Food Category Sources of Saturated Fats in the U.S. Population Ages 2 Years and Older (Figure 2-12)



- **Data Source:** What We Eat in America (WWEI)
A) Food Category analysis

Sodium: Intakes and Limits

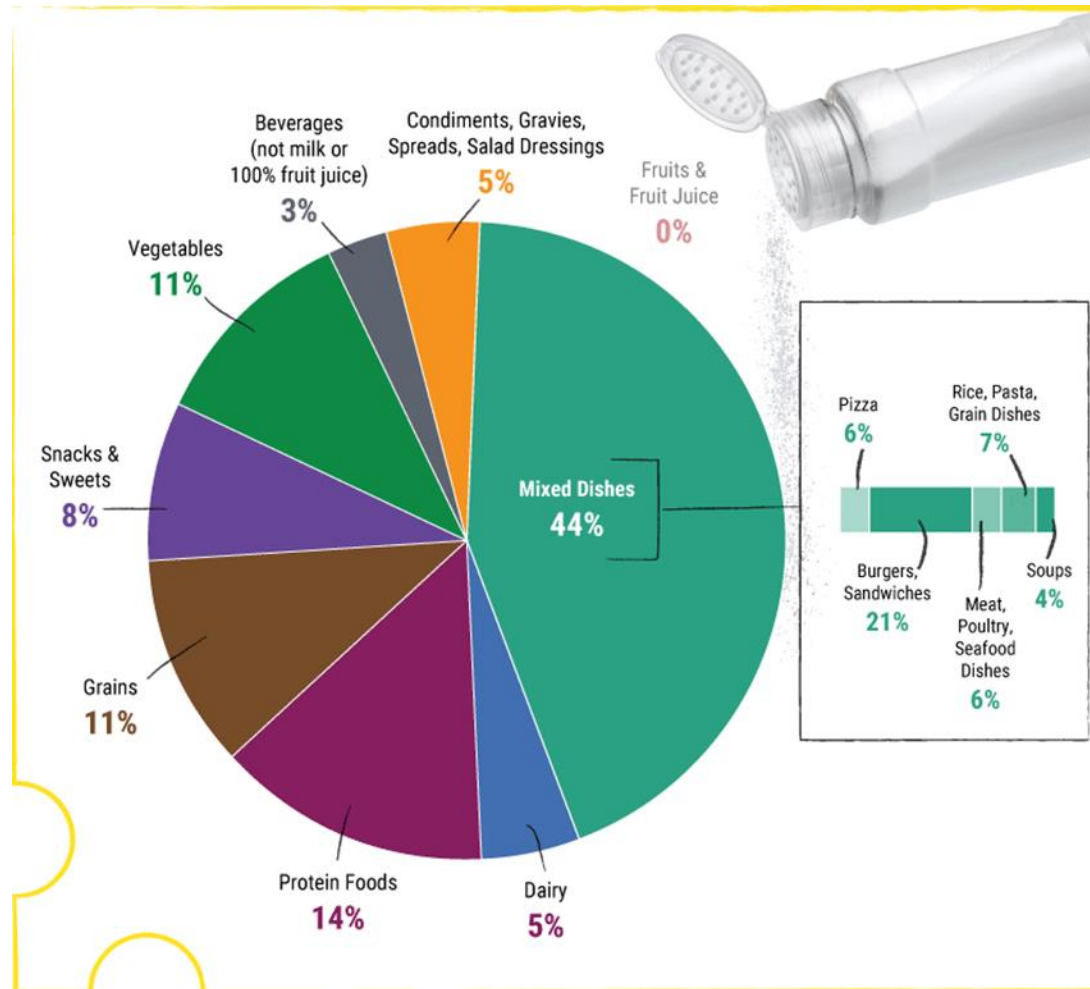
Average Intake of Sodium in Milligrams per Day by Age-Sex Groups, Compared to Tolerable Upper Intake Levels (UL) (Figure 2-13)



- **Data Source:**
- **What We Eat in America, NHANES 2007-2010 for**

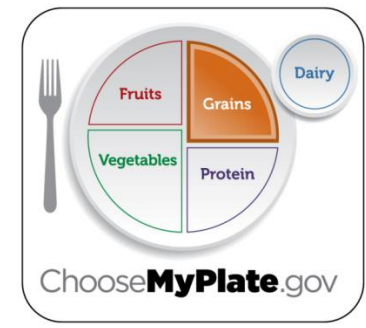
Food Sources of Sodium

Food Category Sources of Sodium in the U.S. Population Ages 2 Years and Older (Figure 2-14)



- **Data Source:** What We Eat in America (WWEIA) Food Category analysis

Grains



Major Nutrient: Carbohydrates, Fiber

Serving size: 1 oz = 1 slice bread, 1 cup dry cereal, ½ cup pasta or rice

Tip: Make at least ½ your grain whole grains

<https://www.myplate.gov/eat-healthy/grains>



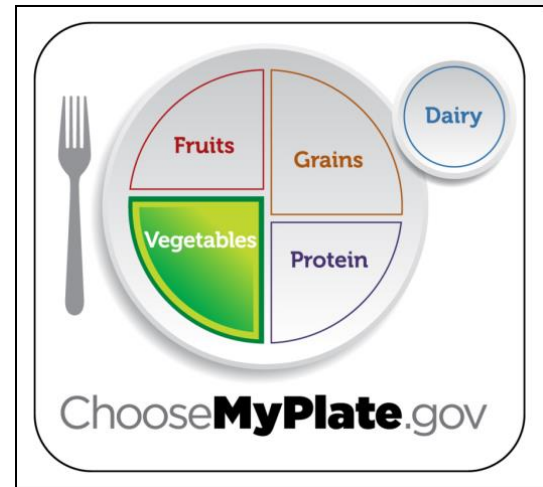
Vegetables

Major Nutrient: Minerals,
Vitamins, Fiber

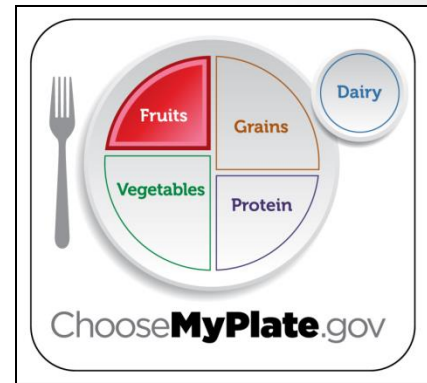
Serving size: $\frac{1}{2}$ cup vegetables = 1 cup leafy vegetables, $\frac{1}{2}$ c.
canned vegetables

Tip: Make half your plate fruits
and vegetables

<https://www.myplate.gov/eat-healthy/vegetables>



Fruits



Major Nutrient: Vitamins, Fiber

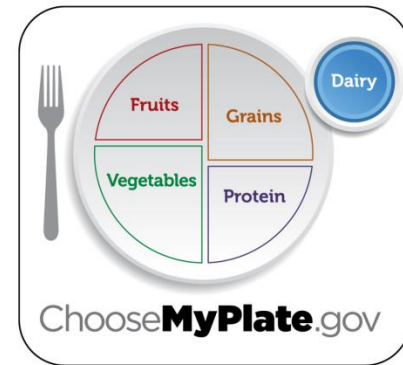
Serving size: 1 cup = 1 medium/small piece of fruit, 1 c. canned, fresh or frozen fruit

Tip: Make half your plate fruits and vegetables

<https://www.myplate.gov/eat-healthy/fruits>



Dairy



Major Nutrient: Minerals, Protein

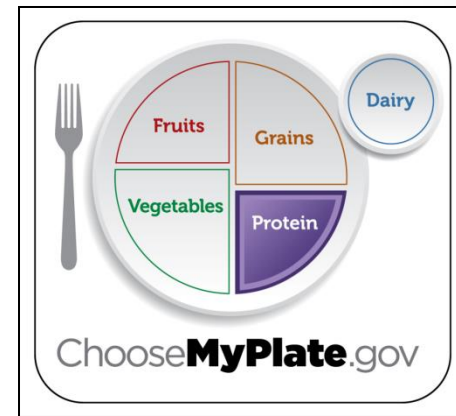
Serving size: 1 cup = 1 ½ oz cheese, 1 cup milk/yogurt/ice cream

Tip: Switch to fat free
or low-fat (1%) milk.

<https://www.myplate.gov/eat-healthy/dairy>



Proteins



Major Nutrient: Protein

Serving size: 1 oz. meat = 1 egg, 1 T peanut butter, ¼ cup cooked beans, ½ oz. nuts or seeds

Tip: choose low fat or lean meats, bake, broil or grill.

Vary protein.

<https://www.myplate.gov/eat-healthy/protein-foods>



Oils

Major Nutrient: Fat

Serving: 3-11 teaspoons

Tip: Choose healthy liquid fats



What triggers your “Eat” response?

- Nervousness/anxiety/stress
- Hormonal fluctuations



- Time of day
- Mood
- Boredom
- Peer/family pressure
- Inattentiveness
- Habit
- Hunger/appetite
- Low self-esteem
- Environment
- Sight and smell of favorite foods

What stops your “Eat” response?

- Acting responsibly in assessing foods
- Practicing stress management
- Breaking the habit
- Remaining active
- Analyzing emotional problems
- Making a conscious effort
- Recognizing true hunger
- Avoiding environment that causes “eat” response



- Selecting alternatives
- Recognizing triggers
- Planning

Lecture and Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>